Math 202 – Slope fields — Spring 2025

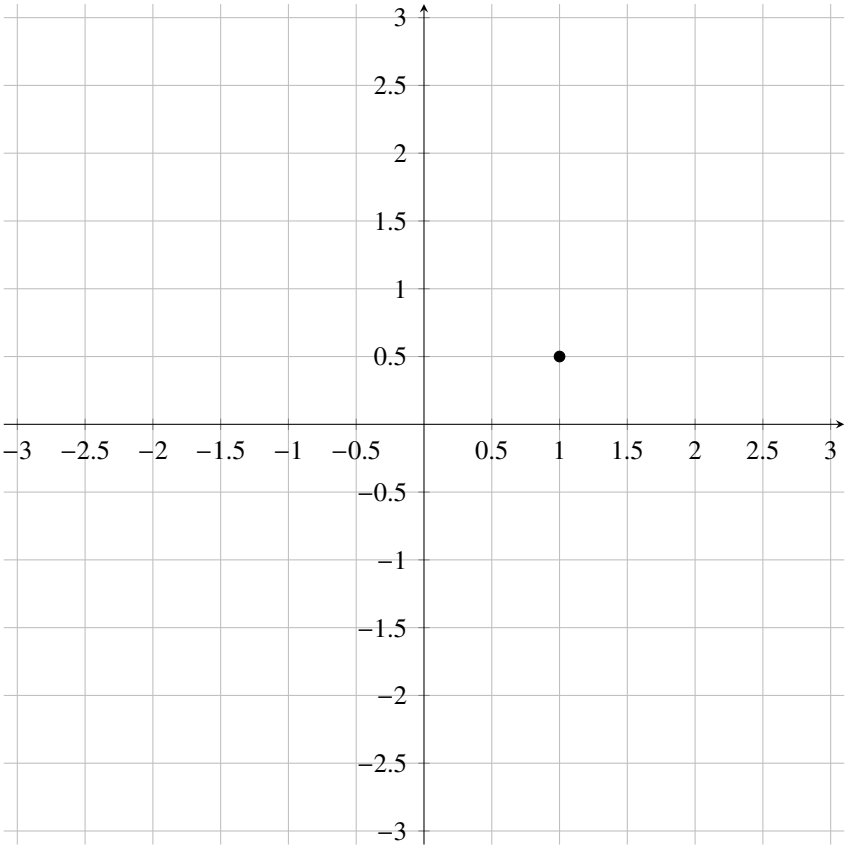
Consider the differential equation $\dfrac{dy}{dx} = -\dfrac{x}{y}$.

a. I've marked one of the corner points of the grid below. What are the coordinates at that point?

b. Plug the coordinates at that point into the differential equation to get out a slope.
(Hint: The answer is -2. Why?)

c. Draw a little tiny tangent line at that point with the slope you got from part b.

d. Repeat this process at each of the corner points in the grid given below, and draw small lines at each point with the slope you find. (You don't need to explicitly evaluate the differential equation at every one of these points, because there are a lot of them, but you should do enough that the pattern becomes clear.)



e. Pick several different initial conditions, and follow the slope field you've generated to sketch the solution curves that result. What do you think the solutions to this differential equation are?